

The ANNNI phase diagram under CNOT-target depolarizing noise

Q-SITE Hacks 2026, Scientific Open Challenge. Benjamin, Dimitri, Yan, Frederick. Code, data and notebook: Scientific Track/team/; every number comes from team/results/analysis/. AI coding assistants were used, as permitted by the organisers.

1. Problem and conventions

We study $H = -\sum_i Z_i Z_{i+1} + \kappa \sum_i Z_i Z_{i+2} - h \sum_i X_i$ with Pauli operators, $J_1 = 1$ and **periodic** rings, for $\kappa \in [0, 1]$ and $h \in [0, 2]$. The phases are identified from longitudinal correlations, not from one-point magnetisations, because the latter vanish in finite symmetric ground states [S07]. We use $S(q) = \frac{1}{N} \sum_{ij} e^{iq(i-j)} \langle Z_i Z_j \rangle$ with the self-terms kept, and the floor-subtracted order parameters $O_F = (S(0) - 1)/(N - 1)$ and $O_A = (S(\pi/2) - 1)/(N/2 - 1)$, which equal 1 for perfect order and 0 for an uncorrelated state. We also use the transverse magnetisation m_x . Validated: $S(0) = N$ (ferro), $S(\pi/2) = N/2$ (antiphase), and $S(q) = 1$ for both $|+\rangle^N$ and $I/2^N$, which m_x separates.

The overlays are approximate thermodynamic references, not finite-ring labels [S10, S13]: the second-order Ising estimate (rationalised so that $h_I(0) = 1$), the lower antiphase-floating fit $1.05(\kappa - \frac{1}{2})$, which is a **Pokrovsky-Talapov** transition [S12], and the upper **KT** fit $1.05\sqrt{(\kappa - \frac{1}{2})(\kappa - 0.1)}$.

2. Methods

Exact reference. Sparse eigsh returns the lowest six eigenpairs on 31×31 grids for $N = 8, 10, 12$, and on fine cuts (21κ values $\times$ $401 h$ values) for all three sizes. Warm-started h sweeps agree with independent cold starts to $1e-13$; degenerate ground spaces are treated as equal mixtures, and fidelity is measured between ground subspaces. The Hamiltonian matches the starter ED and qml.matrix to $7e-15$.

Classifier. The thresholds come from anchor regions fixed by limiting-case physics, not from the overlay curves: ferro at $\kappa \leq 0.2$ and $h \leq 0.2$, antiphase at $\kappa \geq 0.8$ and $h \leq 0.1$, paramagnet at $h \geq 1.8$. This gives $T = 0.541$ for the order parameters and $T_x = 0.490$ for m_x . A point is ferro or antiphase if $\max(O_F, O_A) \geq T$, paramagnet if not but $m_x \geq T_x$, and otherwise *unpolarised*. Labels that flip for thresholds ± 0.1 are marked ambiguous. We also report a threshold-free *relative-dominance* label, $\arg \max(O_F, O_A, m_x^2)$. Under uniform terminal depolarisation all three quantities contract by the same factor a^{2m} with $a = 1 - 4p/3$ [S28], so this label is noise-invariant in that benchmark; any change it shows therefore reveals circuit-specific noise action.

Circuit. We use a Hamiltonian-variational ansatz [S30]: $|+\rangle^N$ followed by $L = N/2 = 4$ layers of $e^{-ib \sum X} e^{-ic \sum ZZ_{nnn}} e^{-ia \sum ZZ_{nn}}$, which gives 12 parameters and preserves translation and parity symmetry. Each ZZ bond compiles to CNOT-RZ-CNOT, for **128 CNOTs** in total. (A hardware-efficient RY/CNOT ring stalled at 1-3 % energy error near the boundaries even at depth 6.) Parameters are optimised at $p=0$ by L-BFGS with exact adjoint gradients, forward and reverse warm-started sweeps and random restarts. Noise: qml.DepolarizingChannel(p) on the target after every CNOT on default.mixed, with the 128 channels verified on the tape; a NumPy density-matrix engine matching default.mixed to $3e-14$ runs the ZNE and response sweeps.

Protocol. Fixed parameters: the same parameters, circuit, grid, thresholds and colour scales at $p = 0, 0.01, 0.05$. The shift $\delta h(\kappa, p) = h_b(p) - h_b(0)$ is taken for the *same circuit*; the preparation bias $h_b(0) - h_b^{\text{ED}}$ is reported separately.

3. Clean results and finite size

At $p=0$ the circuit is an accurate stand-in for the exact ground states. The median relative energy error is $4e-5$ and the median $1 - F$ is $5e-5$ (with F the ground-subspace fidelity). Only 1.5 % of points, all around the multicritical point $\kappa \approx 0.5$, have $F < 0.9$, and the circuit labels agree with ED at 98.9 % of points. The preparation bias on the threshold boundary is at most 0.034 outside $0.42 < \kappa < 0.6$, and 0.066 at $\kappa = 0.57$ next to the multicritical point. Two independent optimiser seeds agree on it to a median of 0.007 (maximum 0.05).

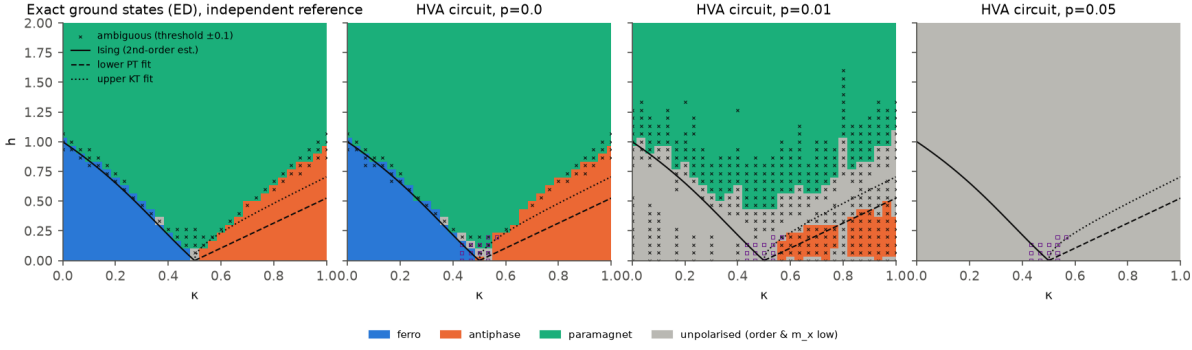


Figure 1. The ED reference and the HVA circuit at $p = 0, 0.01, 0.05$ ($N=8$, PBC, 31×31). Purple squares mark clean preparation fidelity below 0.9; $\times$ marks threshold-ambiguous points.

The **ferro-para** boundary is almost independent of size and tracks the Ising estimate. At $\kappa=0.2$, the fidelity-susceptibility peak moves from 0.616 ($N=8$) to 0.629 ($N=12$), towards the DMRG value 0.639 reported in [S10]; the second-order estimate is 0.653. The **antiphase** boundary drifts strongly with size. At $\kappa=0.75$ the order crossing falls from 0.562 ($N=8$) to 0.432 ($N=12$), approaching the KT fit at 0.423 from above. $N=10$ cannot hold a period-4 pattern and is visibly anomalous: at $\kappa=0.6$ its crossing is 0.397, against 0.244 ($N=8$) and 0.164 ($N=12$). That is a commensurability effect, not physics of the thermodynamic limit.

Floating phase: unresolved. At $N=12$ the half-chain entropy of the pure ground state peaks close to the KT fit ($\kappa=0.75$ and $\kappa=0.9$), and the boundary drifts towards the KT line as N grows. Both are *compatible* with a critical region. However, the dominant q^* stays locked at $\pi/2$ between the PT and KT fits at every size. The only jumps in q^* occur above the KT line, inside the modulated paramagnet [S10], and the momentum resolution ($\Delta q = \pi/6$ at $N=12$) cannot resolve incommensurability. We therefore claim no floating phase [S12, S23]. Label-free k-means on $(S(q)/N, m_x)$ matches the clean rule labels at 99.1 % (consistency, not independent validation).

4. Noise: what shifts, by how much, and why

κ	order	h_T^{ED}	prep. bias	$h_D(0)$	δh_D (0.01)	δh_D (0.05)	h_T at $p=0.01$
0.2	ferro	0.685	-0.000	0.653	+0.027	+0.64 (unreliable)	absent
0.3	ferro	0.487	-0.016	0.469	+0.015	+0.19	absent
0.4	ferro	0.246	+0.000	0.248	+0.003	+0.006	absent
0.7	anti	0.467	+0.018	0.477	-0.001	-0.004	absent
0.8	anti	0.650	+0.024	0.672	+0.003	+0.045	absent
0.9	anti	0.809	+0.004	0.723	+0.000	+0.098	absent

Table 1. h_T is the fixed-threshold crossing and h_D the maximum of $-dO/dh$. The grid step is 0.067; seed spread in δh_D at $p=0.01$ has median 0.011 and maximum 0.038. Full table: *boundary_shifts_circuit_N8_L4.csv*.

$p=0.01$. The order parameters contract to a median of 0.60 of their clean values, against 0.65 if every target hit acted as terminal noise. The contraction depends on the phase. At $h=0$, O_F at $\kappa=0.2$ drops from 1.00 to 0.37, while O_A at $\kappa=0.9$ drops only to 0.59. Under the fixed threshold, **the ferromagnet disappears entirely** (0 % of ED-ferro points keep their label), 46 % of antiphase points and 89 % of paramagnet points survive. Where a threshold crossing survives, it moves **to lower h** by 0.04-0.48, the same sign as the exact first-order prediction $-p \partial_p O / \partial_h O$ but 1.5-4 $\times$ larger, because the response is already nonlinear. The **derivative-peak boundaries barely move**: the median $|\delta h|$ is 0.005 and 28 of 31 rows stay within 0.03 (half a grid step). The exceptions are $\kappa=0.13$ and 0.27 (+0.04) and $\kappa=0.93$ (-0.12). That is what S28 predicts for a near-uniform rescaling, and the relative-dominance map agrees with the clean-ED relative-dominance map at 97 % of points (99 % at $p=0$).

$p=0.05$. The state is essentially maximally mixed: the median purity is 0.0047, against 0.0039 for $I/2^8$. All points are unpolarised under fixed thresholds; in the relative-dominance map the *paramagnet* shrinks:

m_x keeps 16 % of its clean value, so m_x^2 keeps about 2.6 %, while the order parameters keep a median of 8.5 %.

Mechanism: exact channel-insertion sums. We evaluated the first-order response $\partial_p \langle O \rangle = \sum_g \text{Tr}\{O_g \mathcal{L}_t(\rho_g)\}$ location by location [S28]. For the ferro state $d \ln O_F / dp = -99$, for the antiphase $d \ln O_A / dp = -69$, and for m_x in the paramagnet -32 to -48 . The terminal-equivalent value is -43 for pair correlators. Mid-circuit errors, carried through later CNOT-RZ-CNOT gadgets, damage ZZ order more than end-of-circuit errors. Early layers contribute most: for the ferro point, dO_F / dp is -27 from layer 1 and -17 from layer 4. One consistent reading of why the ferromagnet is most fragile: its GHZ-like finite-ring ground state loses all $N(N-1)$ positive correlators in O_F to a single propagated X/Y error, whereas O_A mixes signs and the damage partly cancels. This ranking is specific to this circuit and diagnostic, and reverses the organiser’s hypothesis [S01]. The noise does not change the Hamiltonian’s spectrum, so these are maps of prepared-state diagnostics.

Depth trade-off. We repeated the fixed-parameter experiment for $L = 1-4$ on four κ cuts (depth_tradeoff.json, fig_depth_tradeoff.png). The clean error $\langle |O - O^{\text{ED}}| \rangle_h$ falls with depth, while the noise error grows with it, so the best depth shrinks as p grows. The antiphase is already accurate at $L=2$ (clean error 0.024 at $\kappa=0.8$), and $L=2$ is the best depth there at both $p=0.01$ and $p=0.05$. The ferromagnet needs $L=4$ to be accurate (error at $L=2$: 0.17), and at $p=0.01$ its best depth is $L=3$. So the ferro state needs the full depth and carries the most channels.

5. Error mitigation

Zero-noise extrapolation. We ran Richardson extrapolation on every C_{ij} and $\langle X_i \rangle$ at λp for $\lambda = 1, 2, 3$ (linear scaling), with the $p=0$ truth held out, and rebuilt $S(q)$ before classifying. At **$p=0.01$** this is effective: the median $|\Delta C_{ij}|$ falls from 0.063 to 0.011, $|\Delta m_x|$ from 0.26 to 0.023, and label agreement with the clean circuit rises from 70 % to 98 %. Threshold boundaries come back to within $|\delta h| \leq 0.09$, with a median of 0.03 (for example, $\kappa=0.93$: $-0.48 \rightarrow -0.04$). The cost is a variance-amplification factor $\sum \gamma^2 = 19$. At **$p=0.05$** ZNE fails for both linear and composed-channel scaling: the signal is about 1 % and the extrapolation reaches $p=0.15$.

Parity verification ($P = \prod X_i$) accepts 71 % of the weight at $p=0.01$ and 51 % at $p=0.05$. It lowers the C_{ij} error only from 0.063 to 0.053, because most errors stay inside the even-parity sector.

Noise-aware re-optimisation (separate protocol, $\kappa = 0.2, 0.3, 0.7, 0.8$; 8 h values each). Minimising $\text{Tr}(H\rho_p)$ with exact gradients through the channels lowers the noisy energy by 0.2-1.0 per cut, but the order parameters move by only -0.016 to $+0.028$, and their error against ED is unchanged (for example, $\kappa=0.8$ at $p=0.01$: $0.225 \rightarrow 0.198$). The optimiser gains energy mainly through m_x ($+0.03$ to $+0.09$) by leaving the ground state: at $\kappa=0.2$, $h=0$ the clean energy of the re-optimised parameters is -5.83 , against -6.40 for the true ground state. Energy-optimal noisy states are therefore not diagnostic-optimal (reopt_vs_fixed.json).

6. Conclusions and limitations

The clean diagram is reproduced accurately on 8-12-site rings. For a symmetric circuit accurate enough to represent these ground states (128 CNOTs), **$p=0.01$ already removes the ferromagnet under fixed thresholds while boundary positions defined by derivatives stay put; $p=0.05$ leaves no usable signal.** ZNE recovers the $p=0.01$ map almost fully. Limitations: small N ; overlays are approximate; the $\kappa \approx 0.5$ region is masked for preparation quality; the maps use the fixed-parameter protocol (re-optimisation only on cuts); simulations are exact, with no shot noise.

References (full entries in team/literature/): [S01] organiser handout; [S07] starter-kit audit; [S10] Beccaria, Campostrini, Feo, PRB 73, 052402 (2006); [S12] Cea et al., arXiv:2402.11022 (2024); [S13] Beccaria, Campostrini, Feo, PRB 76, 094410 (2007); [S19] Temme, Bravyi, Gambetta, PRL 119, 180509 (2017); [S21] Bonet-Monroig et al., PRA 98, 062339 (2018); [S23] Sandvik, AIP Conf. Proc. 1297, 135 (2010); [S28] our channel-response derivation; [S30] Wecker, Hastings, Troyer, arXiv:1507.08969 (2015). Tools: PennyLane 0.44.1, NumPy, SciPy, Matplotlib.